

PARTS LIST

Semiconductors:

IC1	- ATmega16 microcontroller
IC2	- 7805, 5V regulator
LED1	- 5mm LED
GLCD1	- 20-pin 128x64 GLCD (KS0108-controller-based)

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 680-ohm
R2	- 10-kilo-ohm
R3	- 470-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 470 μ F, 25V electrolytic
C2	- 0.1 μ F ceramic
C3	- 10 μ F, 16V electrolytic
C4, C5	- 22pF ceramic

Miscellaneous:

CON1	- 4-pin connector
S1	- Tactile switch
X ₁ , 1	- 16MHz crystal oscillator
DC Jack	- DC jack connector
GPS	- GPS-2062 module

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

the navigation data. Port pins PB0 through PB7 of IC1 are connected to data pins D0 through D7 of GLCD1. Port pins PD2 through PD6 are used to provide control signals RS, R/W, EN, CS1 and CS2 to GLCD1, respectively. Switch S1 is used to reset the navigator.

Working of the navigator is simple. Once powered on, the microcontroller stores the initial longitude and latitude data. Then it continuously plots the changing latitude and longitude as you move around (refer Fig. 1). One division change on the screen is equivalent to approximately 30 metres travelled. With the new latitude and longitude position plotted on the screen, you get a clear idea of the direction and path traveled by you. The display also shows various other information such as current latitude, longitude, speed, altitude, date, time and number of satellites the GPS modem can capture.

Software

The software program is written in 'C' language and compiled in WINAVR

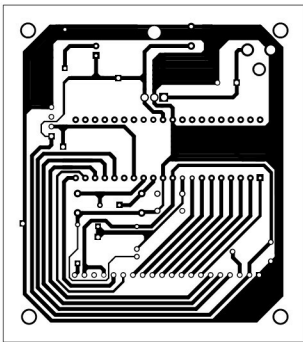


Fig. 3: An actual-size, single-side PCB layout for the GPS navigator

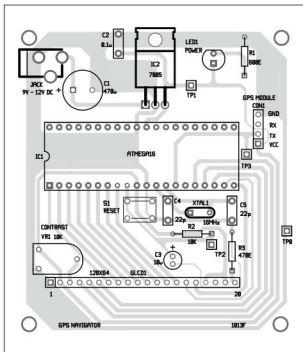


Fig. 4: Component layout for the PCB

Programmers Notepad. WINAVR is a GCC-based compiler for AVR. Burn the program into the MCU using a suitable programmer with FUSE BYTE settings mentioned below:

L Fuse-0XEF

H Fuse-0X99

The GPS receiver module continu-

ously sends data via USART to IC1 at 9600 baud rate. GPS data is initiated by a '\$' sign followed by National Marine Electronics Association (NMEA) output sentences. Details are given in Table I.

The microcontroller continuously captures and stores all bytes for each NMEA output sequence. The total bytes are then partitioned and sampled into a small packet containing information about time, date, longitude, latitude, altitude, speed, etc. These values are regularly updated and displayed on the GLCD.

Construction and testing

An actual-size, single-side PCB for the GPS navigator circuit is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the provided PCB to minimise assembly errors. Use IC base for microcontroller IC1. You can use a 9V/12V adaptor or any other suitable DC source to power the circuit.

To test the circuit for proper functioning, verify 5V power Reset to the system can be checked at TP2. The data transmitted by the GPS modem can be observed at TP3 using an oscilloscope. ●

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